

Class sets and Eisenstein congruences on compact exceptional groups

III. The compact F_4 at level 5

Christopher Younge

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Abstract

We determine the class set of the compact F_4 at the parahoric level attached to the prime 5 (the stabiliser of a sharp-null six-space in $J/5J$): it has 259 classes, 54 over the Albert lattice $J_{\mathbb{Z}}$ and 205 over the exotic lattice J_E , found by uniform and fixed-point sampling of six-spaces with exact stabilisers and exact isomorphism tests, and certified complete by the mass formula, which is met exactly on both halves, together with the independent identity $\sum n_Z/|\text{Stab}| = 713/18$. The Eisenstein primes of this level are therefore 7, 13, 31, 313, 601 (and 691 through $\Theta(\Delta)$). We compute exactly the Iwahori–Hecke generator T_s at 5 on these classes (row sums 5, mass-symmetric, satisfying $(T_s - 5)(T_s + 1) = 0$, spectrum $5^{93}(-1)^{154}$ on the computed part), which bounds the class number of the other maximal parahoric by $93 \leq h' \leq 104$; we prove the lemma that pair-equivalence is stronger than equivalence of intersection lattices; and we record the eigenvalues predicted for the lifts of the classical newforms of level 5 by the dictionaries of Part II, with the primes each is predicted to carry.

1 The level and the main theorem

Let $J = J_{\mathbb{Z}}$ be the integral Albert algebra of Part II and J_E the second class of its genus, $|\text{Aut}(J_{\mathbb{Z}})| = 4,180,377,600$, $|\text{Aut}(J_E)| = 634,023,936$. A six-dimensional subspace $S \subset J/5J$ on which the sharp map vanishes (a *sharp-null six-space*) determines the lattice $\Lambda_S = G_0(S) + 5J$, with $G_0(S)$ the 21-dimensional space of elements whose cross product with S vanishes; the stabiliser of Λ_S in $F_4(\mathbb{Q}_5)$ is a parahoric subgroup Q_5 , and the class set of the compact F_4 at level Q_5 is the set of pairs (J', S) , J' in the genus of J , modulo $F_4(\mathbb{Q})$, i.e.

$$F_4(\mathbb{Q}) \backslash F_4(\mathbb{A}_f) / K = \text{Aut}(J_{\mathbb{Z}}) \backslash \{S \subset J_{\mathbb{Z}}/5J_{\mathbb{Z}}\} \sqcup \text{Aut}(J_E) \backslash \{S \subset J_E/5J_E\},$$

with 38,208,007,656 six-spaces on each side. The mass formula gives $\sum_{J'} \sum_S 1/|\text{Stab}(S)| = 75809539/8294400$ on the $J_{\mathbb{Z}}$ side (Part II) and $5831503/96768$ on the J_E side; a second identity counts, for each S , the number $n_Z(S)$ of its five 5-neighbours isomorphic to $J_{\mathbb{Z}}$: $\sum n_Z/|\text{Stab}| = 713/18$ on the J_E side. Both identities are exact rational numbers, so a list of pairwise distinct orbits whose masses sum to them contains every orbit.

Theorem 1.1. *The class set of the compact F_4 at level Q_5 has 259 classes: 54 over $J_{\mathbb{Z}}$ (Part II) and 205 over J_E . The 205 orbits of $\text{Aut}(J_E)$ on sharp-null six-spaces of $J_E/5J_E$ have stabiliser orders distributed as in Table 1; their masses sum to exactly $5831503/96768$ and $\sum n_Z/|\text{Stab}| = 713/18$ exactly. Consequently the Eisenstein primes of this level (the primes of the numerator of the mass, $2^7 \cdot 3^6 \cdot 7^2 \cdot 13^2 \cdot 31^2 \cdot 313 \cdot 601$ times the Bernoulli factor) are 2, 3, 7, 13, 31, 313, 601 and 691.*

Stab	1	2	4	6	8	12	16	18	24	32
orbits (J_E)	11	66	24	39	13	11	5	1	14	4
Stab	54	64	96	108	168	192	336	1536	6048	
orbits (J_E)	1	2	6	1	1	2	2	1	1	

Table 1: Stabiliser orders of the 205 orbits over J_E . The $J_{\mathbb{Z}}$ side: 8, 14, 4, 1, 7, 4, 1, 1 orbits with stabilisers 2, 4, 6, 8, 12, 24, 36, 48, the twelve large orbits 96, 120, 128, 192^3 , 1152, 2880, 3072^2 , 5184, 9600, and the rare orbit 110,592.

2 Method: finding 205 orbits

Sampling and fingerprints. Uniformly random sharp-null six-spaces (two runs, 41,963 in all) were reduced to the invariants of the intersection lattice $M = \Lambda_S$ (counts of vectors of norm ≤ 4), its stabiliser $|\text{Stab}(S)|$ (the Jordan automorphisms among the automorphisms of M , by the exact test of Part II), and the types and stabilisers of the five lifts. This gives 110 keys, but keys are coarser than orbits: 24 invariant vectors are shared by orbits of different stabilisers, and many keys contain several orbits with the same stabiliser.

Exact splits. For every key, the samples were clustered by the exact isomorphism test (qfism on the intersection lattices followed by the Jordan test on the resulting isometry), 300 samples per key in two runs (by invariant, then by invariant and stabiliser), and a third run for eleven keys whose automorphism closures made the first runs time out. The hit rate per orbit, 702 ± 70 samples per unit of $|\text{Stab}|$, is uniform enough that every key's orbit count is pinned; the top key alone holds ten orbits of stabiliser 2.

The rare orbits. The residual mass after the splits was $367/96768 < 1/192$, so the missing orbits have stabilisers in the hundreds or thousands. Fixed-point sampling (six-spaces invariant under representatives of the 49 conjugacy classes of $\text{Aut}(J_E)$, 298,075 candidates with enlarged symmetry) produced exactly three new invariant vectors, with stabilisers 6048 and 1536 computed and the third too large for the closure; since $1/336 + 1/1536 + 1/6048 = 367/96768$ exactly, the third has stabiliser 336 and the list is complete.

3 The operator T_s at 5: an exact Hecke operator at level Q_5

The level-5 structure (J, Λ_S) is an edge of the Bruhat–Tits building of $F_4(\mathbb{Q}_5)$, between the hyperspecial vertex J and the vertex Λ_S of the other type, so Q_5 is an Iwahori-type subgroup and the parahoric Hecke algebra contains the generator T_s of the reflection through Λ_S : $T_s f(J, S) = \sum_{i=1}^5 f(J_i, S_i)$, the sum over the five other Albert lattices $J_i \supset \Lambda_S$ (the five lifts of S), each with the six-space $S_i = \Lambda_S/5J_i$. Since the double coset has $q = 5$ elements, $(T_s - 5)(T_s + 1) = 0$, and the 5-eigenspace is the space of functions pulled back from the Λ -vertex level, of dimension the class number h' of that maximal parahoric.

Lemma 3.1. *Two pairs (J, S) , (J', S') are in the same class if and only if some $F_4(\mathbb{Q})$ -isometry carries Λ_S onto $\Lambda_{S'}$ and J onto J' . Equivalence of the lattices Λ_S alone is strictly weaker: the five lifts of S have the same Λ_S and are in general different classes.*

This is exactly what the computation of T_s needs: for each class, the five lifts share the representative's intersection lattice, so each lift is identified among the representatives of the same fingerprint by an isometry test (in common ambient coordinates scaled by 80, since the lifts lie in $\frac{1}{5}J$) that is required to map the Albert lattice itself; 1,295 exact identifications in all.

Theorem 3.2. *On 247 of the 258 classes with representatives (all but eleven large-stabiliser classes over $J_{\mathbb{Z}}$, which form a closed block), T_s was computed exactly, with every row sum 5, the mass symmetry $T_{ij}/|\text{Stab}_j| = T_{ji}/|\text{Stab}_i|$ holding on all 55,696 ordered pairs, and $(T_s - 5)(T_s + 1) = 0$ holding on every fully determined row. Its spectrum there is 5 with multiplicity 93 and -1 with multiplicity 154. Hence $93 \leq h' \leq 104$.*

Remark 3.3. A sampled estimate of $T(\omega_1^\vee)(2)$ was also built and is void (Correction 22 in the working notes): the neighbour routine reused from Part II produces the $\omega_4^\vee$ -neighbours, its random Hensel lifting is biased by neighbour type, and, structurally, any feasible sampling quota leaves a noise matrix of spectral norm above every lift eigenvalue. Cheap complete invariants of the orbits do not exist in the obvious form (vector counts to norm 6 separate no more orbits than norm 4), so the exact $T(\omega_1^\vee)(2)$, 36 million exact identifications, remains out of reach; the predictions of the next section stand untested.

4 Predictions

The dictionaries of Part II, re-anchored on 15057, 2331 and 207 at level 3, predict the eigenvalues of $T(\omega_1^\vee)(2)$ on the lifts of the classical newforms of level 5, and the classical Eisenstein congruences of those newforms (verified in their coefficient fields at $q \leq 13$) predict which Eisenstein primes they carry.

Prediction 4.1. The spectrum of $T(\omega_1^\vee)(2)$ at level Q_5 contains 139230 (Eisenstein), 8631 ($\Theta(\Delta)$, carrying 691), 14157.94 and 4868.06 (the $F_4 \times \text{PGL}_2$ lifts of the quadratic pair of weight-12 newforms of level 5, carrying 601), 12285 (the lift of the rational weight-12 newform, $a_2 = 34$, carrying 31), 6668.08 and 1297.92 (the $\text{Sp}_6 \times \text{SL}_2$ lifts of the quadratic pair of weight-8 newforms, carrying 313), 1935 (the 207-type lift of the weight-6 newform, $a_2 = 2$, carrying 31) and 91 (the $\text{Sp}_6 \times \text{SL}_2$ lift of the rational weight-8 newform, $a_2 = -14$, carrying 13). The prime 7 has no classical source in weights 6, 8, 12 and is predicted to fall on a stable form or a G_2 -lift.

The sampled operator (in progress) tests this list; a confirmation would extend the carriers theorem of Part I to a fourth independent level, with the “exceptional” primes (7 here, as $p \mp 1$ on G_2) again on forms with no classical origin.

Data and code

GitHub `chrisyounge-create/Cayley-Agent-Repository`: `f4p5/` (`pathB_E.py`, `orbit_split_reps_E.py`, `fixsample_E.py`, `classify_hunt_E.py`, `reps_ambient.json`, `nbr_sampler.py`, `identify.py`, `opsample.py`, `assemble_operator.py`); all runs as GitHub workflows with artifacts collected into `results/`.